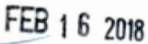


Reference Manual – 83A

PUBLIC HEALTH: Protection and Prevention

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National Park Service
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National Park Service

**PUBLIC HEALTH:
Protection and Prevention
REFERENCE MANUAL- 83A**

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DRINKING WATER POLICY

- 1.1 National Park Service (NPS) unit managers will reduce the risk of waterborne diseases and provide safe drinking water to employees, the visiting public, and park partners by ensuring that drinking water systems are properly operated, maintained, monitored, and deficiencies promptly corrected.
- 1.2 The NPS Office of Public Health (OPH) will assist in providing safe drinking water by conducting a survey of drinking water systems condition, operation and maintenance. The survey is part of a larger onsite assessment classified by NPS OPH as a public health assessment (PHA).
- 1.3 Policy Compliance -- Safe (Potable) drinking water systems will be regulated in accordance with:
 - the Safe Drinking Water Act, as amended (42 U.S.C. 300f¹ et seq.)
 - the Primacy Agency (e.g. the agency designated by Federal law as having oversight responsibility) requirements,
 - this NPS drinking water system policy, whichever is most stringent. Any waivers to this policy must be approved by the NPS OPH Director by completing the form in Appendix 4.0.

Note: If hyperlinks in this document are inactive, please contact your local public health consultant (PHC)². For additional information and reference, the United States Environmental Protection Agency (USEPA) has developed Drinking Water Rule Quick Reference Guides.³

2.0 WATER SYSTEM CLASSIFICATIONS

Water systems are defined as follows or as specified by any changes made in the Safe Drinking Water Act:

- 2.1 **Public Water System (PWS):** A system that, “provides water for human consumption through pipes or other constructed conveyances to at least 15 service connections or serves an average of at least 25 people for at least 60 days a year. A public water system may be publicly or privately owned.”
- 2.2 **Community Water System (CWS):** A public water system that supplies water to the same population year-round.
- 2.3 **Non-Transient Non-Community Water System (NTNC):** A public water system that regularly supplies water to at least 25 of the same people at least six months per year. Examples: Systems serving facilities such as schools or non-residential work

¹ <https://www.gpo.gov/fdsys/pkg/USCODE-2014-title42/pdf/USCODE-2014-title42-chap6A-subchapXII.pdf>

Accessed on 6/15/2017

² http://www.nps.gov/public_health/ Accessed on 6/15/2017

³ <http://www.epa.gov/dwreginfo/drinking-water-rule-quick-reference-guides> Accessed on 6/15/2017

sites where the same people use the water on a regular basis.

- 2.4 **Transient Non-Community Water System (TNC):** Any public water system that serves at least 25 persons per day at least 60 days out of the year. Examples: systems serving campgrounds or other non-residential uses, where people do not remain for long periods of time.
- 2.5 **Non-Public Water Systems (NP):** A water system that does meet one of the above definitions of a public water system. This type of system has less than 15 service connections, or serves less than 25 persons per day on average. Examples could include systems serving ranger stations, individual residences and small campgrounds. The NPS requires NPS managers to follow EPA Drinking Water Standards for individual and/or non-public systems to ensure the health risks associated with contaminants are minimized for all systems.

3.0 PATHOGEN MONITORING

Bacteriological and chemical sampling will be performed in accordance with Federal, State and local laws/regulations along with NPS requirements. All water samples will be tested in laboratories certified by the Primacy Agency. The Total Coliform Rule (TCR) and the Revised Total Coliform Rule (RTCR) (effective April 1, 2016) provide details regarding bacteriological monitoring. Additional information can be found in the following links:

- National Primary Drinking Water Regulations RTCR⁴
- RTCR History⁵
- TCR/RTCR comparison⁶

If there are positive samples, refer to the RTCR for public notification requirements and for water system review requirements.

3.1 Number and Frequency of Samples:

3.1.1 Community Water Systems [CWS]: One sample per month, minimum, or as per Table 1. Samples must be taken at equally spaced intervals each month.

3.1.2 Non-Community Water Systems [NTNC & TNC]: One sample per month, minimum, or as per Table 1. Samples must be taken at equally spaced intervals each month.

3.1.3 Non-Public Water Systems [NP]: One sample per month. The sampling frequency may be changed if authorized in writing by the public health consultant. *Authorization to reduce sampling will only be given after a complete sanitary survey and review of the operational records indicates that reducing the sampling frequency would not result in an increased public health risk to end-users.*

⁴ <https://www.federalregister.gov/articles/2014/02/26/2014-04173/national-primary-drinking-water-regulations-minor-corrections-to-the-revisions-to-the-total-coliform> Accessed on 6/15/2017

⁵ <http://www.epa.gov/dwreginfo/revised-total-coliform-rule-and-total-coliform-rule#rule-history> Accessed on 6/15/2017

⁶ http://www.nps.gov/public_health/ Accessed on 6/15/2017

TABLE 1.0 NPS – TOTAL COLIFORM SAMPLE REQUIREMENT FOR PUBLIC WATER SYSTEMS

Population Served	Minimum Number of Samples per Month	Population Served	Minimum Number of Samples per Month
0 to 1,000	1	8,501 to 12,900	10
1,001 to 2,500	2	12,901 to 17,200	15
2,501 to 3,300	3	17,201 to 21,500	20
3,301 to 4,100	4	21,501 to 25,000	25
4,101 to 4,900	5	25,001 to 33,000	30
4,901 to 5,800	6	33,001 to 41,000	40
5,801 to 6,700	7	41,001 to 50,000	50
6,701 to 7,600	8	50,001 to 59,000	60
7,601 to 8,500	9	For populations > 59,000 see ⁷ (pg 10361)	

The population served (**p**) is defined as the sum of the residents (**r**) and the average daily transient population (total number of transients (**t**) served per month divided by the number of days (**d**) of the month during which the transients were served), i.e., $p = r + t/d$. (If the transient population changes significantly from month to month, utilize information from previous years of operation to calculate the average daily transient population for the current month). If there are questions regarding the transient population served consult with the primacy agency and PHC.

Note: the local, State or County jurisdiction may require an increased sampling protocol based on local regulations or a record of poor test results.

3.2 Special Sampling Requirements:

3.2.1 Seasonal systems: Seasonal systems must obtain two consecutive, **negative** samples prior to utilization of the system. Samples may be collected on the same day, but not at the same time and must be collected at different sampling sites according to the site-sampling plan, if applicable. Example shut-down and start-up guides are located in appendix 7.0 and on the NPS OPH website⁸. Also the RTCR requires that the primacy agency establishes a start-up procedure. Therefore, if the primacy agency has a start-up procedure that is required, utilize that and provide a copy to your PHC. Propylene glycol or other anti-freeze type liquids are not allowed unless approved through waiver process through regional PHC.

⁷ <https://www.gpo.gov/fdsys/pkg/FR-2013-02-13/pdf/2012-31205.pdf> Accessed on 6/15/2017

⁸ http://www.nps.gov/public_health/eh/dw.htm Accessed on 6/15/2017

- 3.2.2 Source water monitoring: Some Primacy Agencies require periodic source (raw) water monitoring. Sampling of source water must be in accordance with the Primacy Agency requirements.
- 3.2.3 Municipal supplies: National Park Service operated distribution systems serviced by municipal systems should be included in the municipality's bacteriological sampling programs whenever feasible. When the municipality does not monitor a distribution system, the PHC should be contacted to determine if and when the system requires sampling.
- 3.3 Site-sampling plan: Each system must have a written site-sampling plan (examples^{9,10}) available for review by the Primacy Agency, and/or the PHC. All bacteriological samples must be collected from identified sampling sites. An example plan can be found in appendix 5.0 or the NPS OPH website at: https://www.nps.gov/public_health/. If links are inactive, please conduct a search in your web browser for, National Park Service Office of Public Health Drinking Water.
- 3.4 Positive samples: All positive total coliform samples require that the system operator obtain repeat samples in accordance with the Primacy Agency and, at a minimum, the following:
- 3.4.1 Public systems: Three repeat samples must be collected within 24 hours of laboratory notification of the positive result. (See Ground Water Rule (GWR) and RTCR for details on resampling and contact your PHC. Details can be found on page 17 & 18 of 40 CFR Parts 141 and 142¹¹.) **NOTE**: This includes public non-community systems.
- 3.4.2 Non-public systems: Two repeat samples must be collected and analyzed. One sample must be collected at the site of the positive sample. If possible, three samples should be collected, one at the positive sample location, one upstream, and one downstream. For example: on a residential system, one should be taken inside the residence and one should be taken outside the residence. Notify your PHC of initial positive and repeat samples results upon discovery.
- 3.5 Non-compliance - Public Systems: **If any repeat samples are positive (TC+)**, the park must notify the Primacy Agency and the PHC within 24 hours following receipt of notification of sample results. The Primacy Agency and the PHC must be contacted for assistance in determining the source of the contamination and in implementing corrective actions.
- 3.6 Non-compliance - Non-Public Systems: **If any repeat samples are positive (TC+)**, the park must notify the PHC within 24 hours following receipt of notification of sample results. The PHC must be contacted for assistance in determining the source of the contamination and in implementing corrective actions.

⁹ http://www2.epa.gov/sites/production/files/2015-04/documents/rtrc_siting_plan_template.pdf Accessed on 6/15/2017

¹⁰ http://www.nps.gov/public_health/eh/dw.htm Accessed on 6/15/2017

¹¹ <https://www.gpo.gov/fdsys/pkg/FR-2013-02-13/pdf/2012-31205.pdf> Accessed on 6/15/2017

4.0 CHEMICAL MONITORING

Monitoring requirements are as shown in Table 5: Summary of Parameters and Testing Regime. Test results will be submitted to the Primacy Agency and PHC upon receipt. There are several federal rules, and state primacy agencies may establish additional monitoring requirements. Many of the rules are described below and in Table 5, however verify your particular sampling requirements with the Primacy Agency. Many states follow 40 CFR 141.23 – “Inorganic chemical sampling and analytical requirements” monitoring as listed below:

- Groundwater systems shall take a minimum of one sample at every entry point to the distribution system, which is representative of each well after treatment (hereafter called a sampling point) beginning in the initial compliance period. The system shall take each sample at the same sampling point unless conditions make another sampling point more representative of each source or treatment plant.
- Surface water systems shall take a minimum of one sample at every entry point to the distribution system after any application of treatment or in the distribution system at a point, which is representative of each source after treatment (hereafter called a sampling point) beginning in the initial compliance period. The system shall take each sample at the same sampling point unless conditions make another sampling point more representative of each source or treatment plant.

The EPA National Primary Drinking Water Regulations must be met for drinking water systems. Details about contaminants¹² can be found on Fact Sheets¹³ developed by the EPA.

- 4.1 Primary inorganics, secondary contaminants, and general minerals: Primary inorganic, secondary contaminants, and general mineral analyses must be conducted to meet regulatory requirements and/or to determine the most feasible treatment methodologies to provide water of satisfactory quality. Samples shall be collected from the source before treatment. NOTE – Nitrates are required annually for all sources. Disinfection byproduct sampling is performed at locations within the distribution system, post treatment.

For more information on inorganic chemicals, specific rule information, compliance dates, and regulatory updates, see EPA Drinking Water contaminants and standards¹⁴.

4.2 Organics:

4.2.1 Pesticides, Herbicides & Polychlorinated Biphenyls (PCB)'s: All systems with community and non-transient users must be sampled annually unless a State or Primacy Agency waiver has been issued. All transient non-community and non-public systems must be sampled one time. Additional sampling requirements are dependent upon the

¹² <http://www.epa.gov/dwstandardsregulations> Accessed on 6/15/2017

¹³ <https://safewater.zendesk.com/hc/en-us/categories/201454937> Accessed on 6/15/2017

¹⁴ <https://safewater.zendesk.com/hc/en-us/categories/201454957> Accessed on 6/15/2017

initial sampling results and source vulnerability. Samples should be collected after treatment and prior to entry to the distribution system.

4.2.2 **Volatile Organic Chemicals (VOC)**: There are fifty-five VOC's that have Maximum Contaminant Level (MCL's) (regulated or unregulated) that must be analyzed. Samples should be collected after treatment and prior to entry to the distribution system.

4.2.2.1 **Groundwater Supplies**. Public community and public non-transient non-community systems served by groundwater supplies should have collected a sample from each entry point to the distribution system for an initial analysis by **December 1990**. A single routine sample must be collected at three-year intervals thereafter, unless the Primacy Agency has issued a waiver.

4.2.2.2 **Surface Water Systems**. The initial sampling for public community and public non-transient non-community systems served by surface sources consists of four quarterly samples to be collected over a one-year period. The first sample should have been collected by **December 1990**. Sampling must be repeated at three-year intervals thereafter, unless the Primacy Agency has issued a waiver.

4.2.3 **All Other Public Systems** (ground or surface water). These systems should be sampled one time for baseline information. Contact the PHC for additional information and guidance.

4.2.4 **Non-public Systems**. These systems should be sampled in special situations only. Contact the PHC for additional information and guidance.

For more information on organic chemicals, specific rule information, compliance dates, and regulatory updates, search EPA Drinking Water contaminants and standards¹⁵.

Note: If the primacy agency grants a waiver for organic chemical monitoring for a public system, the PHC may also grant a waiver for those non-public systems not regulated by the State but lie within the same aquifer.

4.3 **Disinfectants/Disinfection Byproducts**: This rule is intended to balance the use of disinfectants with the potential health effects from long term exposure to byproducts formed by the reaction of disinfectants with natural organic and inorganic matter in drinking water.

4.3.1 This rule applies to all community and non-transient non-community water systems that add a chemical disinfectant to the water in any part of the treatment process. This includes surface water systems, groundwater systems under the direct influence of surface water, and groundwater systems.

4.3.2 **Surface Water and Ground Water Under the Direct Influence of Surface Water**:

¹⁵ <https://safewater.zendesk.com/hc/en-us/categories/201454957> Accessed on 6/15/2017

All surface water sources and any groundwater sources under the direct influence (GWUDI) of surface water, as determined by the Primacy Agency for public systems will be provided with approved treatment processes. Non-public surface water sources and groundwater sources under the direct influence of (GWUDI) surface water, as determined by the PHC, will be provided with approved treatment process. For additional detail see EPA GWUDI regulations.

4.3.3 Byproducts to be monitored:

- Total trihalomethanes (TTHM): The sum of chloroform, bromoform, bromodichloromethane, and dibromochloromethane.
- Haloacetic acids (HAA5): The sum of monochloroacetic acid, dichloroacetic acid, trichloroacetic acid, monobromoacetic acid, and dibromoacetic acid.
- Bromate: For those systems using ozone for disinfection.
- Chlorite: For those systems using chlorine dioxide for disinfection.

4.3.4 Residual Disinfectants to be monitored:

- Chlorine: for those systems using chlorine.
- Chloramine: for those systems using chloramine.
- Chlorine dioxide: for those systems using chlorine dioxide.

4.3.5 Monitoring requirements: Community and Non-Transient Non-Community water systems. Sample GWUDI sources with the same frequency as surface water sources.

- Large surface water systems (>10,000 persons/day): monitor 4 samples/plant/quarter for TTHMs and HAA5's.
- Small surface water systems (<10,000 persons/day): monitor 1 sample/plant/quarter for TTHMs and HAA5's.
- Very small surface water systems - serving less than 500 people each day: monitor 1 sample/plant/year for TTHMs and HAA5s.
- Large ground water systems - serving more than 10,000 people each day: monitor 1 sample/plant/quarter for TTHMs and HAA5s.
- Small ground water systems - serving less than 10,000 people each day: monitor 1 sample/plant/year for TTHMs and HAA5s.

4.3.6 Monitoring requirements: For transient and non-public water systems (also refer to section 7.3 in this document).

- Transient surface water and ground water systems: As required by PHC.*
- Non-public surface water and ground water systems: As required by PHC.*

**Note: Additional monitoring may be requested based on scientific reasoning to include water source type, contact time, total organic carbon (TOC) concentrations, past testing results, and/or other water quality factors.*

4.4 Chemical Monitoring Waivers

Potable drinking water systems connected to municipal supplies are not required to perform chemical monitoring if current analyses for the area are available from the municipality. If current analyses are not available, the Park may be directed by the PHC to perform the analyses.

The monitoring requirements in this section may be waived for non-public and/or unregulated systems if authorized, in writing, by the PHC. This authorization will only be given after a complete sanitary survey of the system and review of the operational records indicates a reduced sampling frequency or waiver would not increase risk to end-users.

For specific rule information, compliance dates, and regulatory updates, go to:

EPA Chemical Contaminant Rules¹⁶

5.0 LEAD AND COPPER

On June 7, 1991, final national primary drinking water regulations for lead and copper were adopted for public-community and public non-transient non-community water systems. The rule establishes maximum contaminant level goals (MCLG), and action levels for both lead and copper. It also sets forth treatment technique requirements and specifies monitoring requirements, analytical methods, public notification requirements, record keeping and reporting requirements, and compliance schedules. An EPA public education program is also required if an action level is exceeded. All community water systems (CWS) and non-transient non-community (NTNC) water systems are subject to the Lead and Copper Rule (LCR)¹⁷.

To comply with the rule and in order to assess for and reduce the health risks associated with lead and copper, parks will:

5.1 Surveillance for All NPS Water Systems:

5.1.2 All CWSs and NTNCs must collect first-draw samples at taps in homes/buildings

¹⁶ <http://www.epa.gov/dwreginfo/chemical-contaminant-rules> Accessed on 6/15/2017

¹⁷ <http://www.epa.gov/dwreginfo/lead-and-copper-rule> Accessed on 6/15/2017

that are at high risk of Lead (Pb)/ Copper(Cu) contamination as identified in the Code of Federal Regulations (40 CFR 141.86(a)). If parks have not completed this, it should be conducted immediately.

5.1.3 All non-public water systems should take first draw tap samples (one liter) at each Park Service owned non-public system and have them analyzed for lead and copper. If one or more first draw tap samples are elevated for lead or copper (>0.015 mg/l, >1.3 mg/l respectively), contact your PHC for further directives.

5.1.4 Assure that lead solder or flux is no longer used in domestic water plumbing.

5.1.5 Replace drinking fountains listed by EPA under the Lead Contamination Control Act.

5.2 Monitoring:

5.2.1 All CWSs and NTNCs must collect the number of samples as shown in Table 2.0 and 2.1 below, and systems must conduct monitoring every six months unless they qualify for reduced monitoring (Table 2.1). Water quality parameter (WQP) samples at entry points to distribution system (EPTDS) are collected every 6 months prior to any corrosion control treatment.

5.2.2 For TNC and Non-public systems, NPS operators should contact their regional PHC for directives on Pb and Cu monitoring.

5.2.3 Regional PHCs cannot contradict the primacy agency.

TABLE 2.0 LEAD AND COPPER TAP AND WQP TAP MONITORING (DOES NOT APPLY TO EPTDS WQP MONITORING.)¹⁸

Size Category	System Size (People)	Number of Pb/Cu Tap Sample Sites ³		Number of WQP Tap Sample Sites ⁴	
		Standard	Reduced	Standard	Reduced
Large	> 100K	100	50	25	10
	50,001 - 100K	60	30	10	7
Medium	10,001 - 50K	60	30	10	7
	3,301 - 10K	40	20	3	3
Small	501 - 3,300	20	10	2	2
	101 - 500	10	5	1	1
	≤ 100	5	5	1	1

³ With written State approval, PWSs can collect < 5 samples if all taps used for human consumption are sampled.

⁴ Two WQP tap samples are collected at each sampling site.

¹⁸ <http://nepis.epa.gov/Exe/ZyPDF.cgi?Dockey=60001N8P.txt> Accessed on 6/15/2017

TABLE 2.1 CRITERIA FOR REDUCED Pb/Cu TAP MONITORING	
Annual	1 PWS serves $\leq 50,000$ people and is $\leq$ both established action levels (ALs) for 2 consecutive 6-month monitoring periods; or 2 Any PWS that meets optimal WQPs (OWQPs) and is $\leq$ Pb AL for 2 consecutive 6-month monitoring periods.
Triennial	1. PWS serves $\leq 50,000$ people and is $\leq$ both ALs for 3 consecutive years of monitoring; or 2. Any PWS that meets OWQP specifications and is $\leq$ Pb AL for 3 consecutive years of monitoring; or 3. Any PWS with 90th percentile Pb and Cu levels ≤ 0.005 mg/L and ≤ 0.65 mg/L, respectively, for 2 consecutive 6-month monitoring periods (i.e., accelerated reduced
Every 9 years	PWS serves $\leq 3,300$ people and meets monitoring waiver criteria found at 40 CFR 141.86(g).

6.0 RADIONUCLIDES¹⁹

In 2000, EPA revised the radionuclides regulation²⁰, which had been in effect since 1977. The revisions required new monitoring provisions to ensure that all customers of community water systems will receive water that meets the Maximum Contaminant Levels for radionuclides in drinking water. EPA also issued a standard for uranium, as required by the 1986 amendments to the Safe Drinking Water Act. The current standards are: combined radium 226/228 of 5 pCi/L; a gross alpha standard for all alphas of 15 pCi/L (not including radon and uranium); a combined standard of 4 mrem/year for beta emitters. The new MCL for uranium is 30 μ g/L. Radionuclide sample should be taken before treatment.

Community water systems (CWSs), are required to meet the final MCLs and to meet the requirements for monitoring and reporting.

Non-transient, non-community water systems (NTNCWSs) with a ground water source will be tested one time, unless waived by the PHC.

For transient non-community (TNC) and non-public (NP) systems, NPS operators should contact their regional PHC for directives on radionuclides testing.

7.0 WATER TREATMENT

7.1 Filtration:

7.1.1 All surface water²¹ sources and any groundwater sources under the direct influence of surface water must be provided with approved filtration.

7.1.2 Compliance schedule for provision of filtration: For groundwater supplies that are determined to be under the influence of surface water, filtration must be in place and operational 18 months after the Primacy Agency officially issues the determination.

7.1.3 Parks with groundwater sources clearly under the direct influence of surface water should contact the PHC for assistance in developing a schedule to provide filtration or developing a new groundwater source.

¹⁹ <http://www.epa.gov/radiation/radionuclides> Accessed on 6/15/17

²⁰ <http://www.epa.gov/dwreginfo/radionuclides-rule> Accessed on 6/15/2017

²¹ https://www.epa.gov/sites/production/files/documents/SWTR_Fact_Sheet.pdf Accessed on 6/15/2017

7.1.4 In those cases where it is not certain if a groundwater source is under the influence of surface water, parks should be aware that the criteria for making such determinations have not been developed by all Primacy Agencies. When these criteria are developed, parks should contact the PHC for assistance in developing and initiating a modification schedule. Microscopic Particulate Analysis (MPA) can be used to help assess whether a groundwater source is under the direct influence (GWUDI) of surface water.

7.1.5 On December 16, 1998, the EPA finalized an Interim Enhanced Surface Water Treatment Rule (IESWT)²², which became effective **February 16, 1999**. This rule is intended to improve control of microbial pathogens, including specifically the protozoan *Cryptosporidium*, by enhancing physical removal efficiencies in drinking water, and addresses risk trade-offs with disinfection byproducts.

7.1.6 In January 2004, the EPA Long Term 1 Enhanced Surface Water Treatment Rule (LT1ESWTR) became effective. This rule follows the IESWTR and regulates systems of less than 10,000 people. This rule includes the compliance with Stage 1 Disinfectants/Disinfection Byproducts Rule. These new rules affect public and non-public surface water treatment plants, including ground water systems under the influence of surface water.

7.1.7 In December 2005, the EPA Long Term 2 Enhanced Surface Water Treatment Rule (LT2ESWTR)²³ and the Stage 2 Disinfection Byproduct Rule²⁴ became effective. The Long Term 2 Enhanced Surface Water Treatment Rule and the Stage 2 Disinfection Byproducts Rule are the second phase of rules required by Congress. These rules strengthen protection against microbial contaminants, especially *Cryptosporidium*, and at the same time, reduce potential health risks of disinfection byproducts. The new regulations apply to public water systems that use surface water or ground water under the influence of surface water.

Under the LT2ESWTR, systems will monitor their water sources to determine treatment requirements. This monitoring includes an initial two years of monthly sampling for *Cryptosporidium*. To reduce monitoring costs, small filtered water systems will first monitor for *E. coli*—bacterium which is less expensive to analyze than *Cryptosporidium*—and will monitor for *Cryptosporidium* only if their *E. coli* results exceed specified concentration levels.

Monitoring start dates are staggered by system size, with larger systems being monitored before smaller systems. Systems must conduct a second round of monitoring six years after completing the initial round to determine if source water conditions have changed significantly. Systems may use (grandfather) previously collected data in lieu of

²² <https://www.epa.gov/dwreginfo/implementation-guidance-appendices-surface-water-treatment-rules> Accessed on 6/15/2017

²³ <https://www.epa.gov/dwreginfo/long-term-2-enhanced-surface-water-treatment-rule-documents> 6/15/2017

²⁴ <http://www.epa.gov/dwreginfo/stage-1-and-stage-2-disinfectants-and-disinfection-byproducts-rules> 6/15/2017

conducting new monitoring, and systems are not required to monitor if they provide the maximum level of treatment required under the rule.

Cryptosporidium treatment: Filtered water systems will be classified in one of four treatment categories (bins) based on their monitoring results. The majority of systems will be classified in the lowest treatment bin, which carries no additional treatment requirements. Systems classified in higher treatment bins must provide 90 to 99.7 percent (1.0 to 2.5-log) additional treatment for *Cryptosporidium*. Systems will select from a wide range of treatment and management strategies in the "microbial toolbox" to meet their additional treatment requirements. All unfiltered water systems must provide at least 99 or 99.9 percent (2 or 3-log) inactivation of *Cryptosporidium*, depending on the results of their monitoring. These *Cryptosporidium* treatment requirements reflect consensus recommendations of the Stage 2 Microbial and Disinfection Byproducts Federal Advisory Committee.

Other requirements: Systems that store treated water in open reservoirs must either cover the reservoir or treat the reservoir discharge to inactivate 4-log virus, 3-log *Giardia lamblia*, and 2-log *Cryptosporidium*. These requirements are necessary to protect against the contamination of water that occurs in open reservoirs. In addition, system operators must review their current level of microbial treatment before making a significant change in their disinfection practice. This review will assist to protect against microbial pathogens as they take steps to reduce the formation of disinfection byproducts under the Stage 2 Disinfection Byproducts Rule, which EPA is finalizing along with the LT2ESWTR.

The Stage 2 Disinfection Byproducts Rule (DBPR) strengthens public health protection for customers by tightening compliance monitoring requirements for two groups of Disinfection Byproducts (DBPs), trihalomethanes (TTHM) and haloacetic acids (HAA5). (Also refer to section 4.3 in this document.) The rule targets systems with the greatest risk and builds incrementally on existing rules. This regulation will reduce DBP exposure and related potential health risks and provide more equitable public health protection.

Under the Stage 2 DBPR, systems will conduct an evaluation of their distribution systems, known as an Initial Distribution System Evaluation (IDSE), to identify the locations with high DBP concentrations. These locations will then be used by the systems as the sampling sites for Stage 2 DBPR compliance monitoring.

Compliance with the maximum contaminant levels for two groups of DBPs (TTHM and HAA5) will be calculated for each monitoring location in the distribution system. This approach, referred to as the locational running annual average (LRAA), differs from current requirements, which determine compliance by calculating the running annual average of samples from all monitoring locations across the system.

The Stage 2 DBPR also requires each system to determine if they have exceeded an operational evaluation level, which is identified using their compliance monitoring results. The operational evaluation level provides an early warning of potential future maximum contaminant level (MCL) violations, which allows the system to take proactive steps to remain in compliance. A system that exceeds an operational evaluation level is required to review their operational practices and submit a report to their state that

identifies actions that may be taken to mitigate future high DBP levels, particularly those that may jeopardize their compliance with the DBP MCLs.

Entities potentially regulated by the Stage 2 DBPR are community and nontransient noncommunity water systems that produce and/or deliver water that is treated with a primary or residual disinfectant other than ultraviolet light.

7.1.8 The EPA Ground Water Rule (GWR)²⁵ reduces the risk of exposure to fecal contamination that may be present in public water systems that use ground water sources. The rule establishes a risk-targeted strategy to identify ground water systems that are at high risk for fecal contamination. The GWR²⁶ also specifies when corrective action (which may include disinfection) is required to protect consumers from bacteria and viruses if they consume water from ground water systems. The rule addresses risks through a risk-targeting approach that relies on four major components:

7.1.8.1 Periodic sanitary surveys of ground water systems that require the evaluation of eight critical elements and the identification of significant deficiencies (e.g., a well located near a leaking septic system).

7.1.8.2 Source water monitoring to test for the presence of *E. coli*, *enterococci*, or *coliphage* in the sample. There are two monitoring provisions:

- *Triggered monitoring* for systems that do not already provide treatment that achieves at least 99.99 percent (4-log) inactivation or removal of viruses and that have a total coliform-positive routine sample under Total Coliform Rule sampling in the distribution system.
- *Assessment monitoring* - As a complement to triggered monitoring, a State has the option to require systems, at any time, to conduct source water assessment monitoring to help identify high risk systems.

7.1.8.3 Corrective actions required for any system with a significant deficiency or source water fecal contamination. The system must implement one or more of the following correction action options:

- correct all significant deficiencies,
- eliminate the source of contamination,
- provide an alternate source of water, or
- provide treatment which reliably achieves 99.99 percent (4-log) inactivation or removal of viruses.

7.1.8.4 Compliance monitoring to ensure that treatment technology installed to treat drinking water reliably achieves at least 99.99 percent (4-log) inactivation or removal of viruses.

²⁵ <http://www.epa.gov/dwreginfo/ground-water-rule> Accessed on 6/15/2017

²⁶ <http://nepis.epa.gov/Exe/ZyPDF.cgi?Dockey=P1001KJK.txt> Accessed on 6/15/2017

7.1.9 Filter Backwash Recycling²⁷: This rule requires systems that recycle to return specific recycle flows through all processes of the system's existing conventional or direct filtration system or at an alternate location approved by the state (66 FR 31086, June 8, 2001, Vol. 66²⁸).

7.2 Turbidity Monitoring:

7.2.1 All water systems requiring filtration under the Surface Water Treatment Rule, Interim Enhanced Surface Water Treatment Rule, Long Term 1 Enhanced Surface Water Treatment Rule and Long Term 2 Enhanced Surface Water Treatment Rule (7.1.6) must be analyzed for turbidity. The unit of measure for turbidity is the Nephelometric Turbidity Unit (NTU). The MCL is 1.0 NTU for diatomaceous earth and slow sand filtration and 0.3 NTU for conventional and direct filtration. The MCL for "other technology" filtration such as bag filtration is 1.0 NTU unless a more stringent requirement is established by the Primacy Agency. For systems that use conventional or direct filtration, at no time can turbidity (cloudiness of water) go higher than 1 Nephelometric Turbidity Unit (NTU), and samples for turbidity must be less than or equal to 0.3 NTUs in at least 95 percent of the samples in any month. Systems that use filtration other than the conventional or direct filtration must follow state limits, which must include turbidity at no time exceeding 5 NTUs. Systems serving 500 or fewer persons per day must collect one grab sample per day. Systems serving more than 500 persons per day must collect one grab sample every four hours when water is served to the Public or continuous monitoring may be substituted for grab sampling.

7.2.2 The PHC must be informed of any violation. The issuance and content of public notification must be determined in consultation with the Primacy Agency and/or PHC.

7.3 Disinfection:

All public drinking water systems will be continuously disinfected with NSF/ANSI 60 Standard 60 for Drinking Water Treatment Chemicals. Sodium hypochlorite is the preferred disinfectant: primary disinfectants such as ozone and ultraviolet irradiation do not provide a residual and therefore an acceptable secondary disinfection will be required. Primary disinfection may be used for point of use treatment with low risk of recontamination by the user. Approval by the Primary Agency (if applicable) and the PHC is required. The PHC may specifically exempt **non-public** systems of primary and secondary disinfection after a complete sanitary survey of the system is made.

It is recommended that disinfection of groundwater systems achieve 4-log virus inactivation if not achieved by other methods (e.g., reverse osmosis (RO), ozone, ultraviolet). New construction should be designed to provide 4-log virus inactivation as specified by PHC. Drinking water systems and sources with several positive bacterial samples may need to be replaced or require 4- log virus inactivation. Parks with surface water systems must monitor the entry point to the distribution system

²⁷ <http://nepis.epa.gov/Exe/ZyPDF.cgi?Dockey=200025V5.txt> Accessed on 6/15/2017

²⁸ <https://www.gpo.gov/fdsys/pkg/FR-2001-06-08/pdf/01-13776.pdf> Accessed on 6/15/2017

for the presence of a chlorine residual. The minimum chlorine residual is 0.2 mg/l or a higher level determined using CT calculations. If the residual drops below 0.2 mg/l (or the value required for minimum CT, whichever is greater), it must be restored within four hours. Whenever the residual falls below the required value, the park must notify the Primacy Agency and the PHC as soon as possible but no later than the end of the next business day. For systems serving 3,300 persons or more, the chlorine residual must be monitored on a continuous basis. Systems serving fewer than 3,301 persons can take grab samples in lieu of continuous monitoring at the following frequencies:

System Population	Samples/day
< 500	1
501-1000	2
1001-2500	3
2501-3300	4

Acceptable secondary disinfection methods are those which provide a measurable disinfectant residual in the distribution system.

7.3.1 Distribution System Residual: All potable drinking water systems will be continuously disinfected, the PHC may specifically exempt ***non-public*** systems after a complete sanitary survey of the system is made. A disinfectant residual must be maintained (e.g. minimum 0.2 mg/L free chlorine) throughout the distribution system for all continuously disinfected drinking water systems. If disinfectants other than chlorine are used, the PHC or Primacy Agency must be contacted for residual levels required to be maintained. The absence of a residual indicates either an equipment failure or the presence of contamination in the system. If an equipment failure is not the cause, the park must contact the PHC or the Primacy Agency for assistance.

7.3.2 Routine Monitoring: All drinking water distribution systems must be monitored for the presence of an acceptable residual. For those park operated, public drinking water systems utilizing a chemical disinfectant, one (1) disinfectant residual sample will be measured and recorded per day from representative points unless advised otherwise by the PHC (also shown in table 5.0). Parks operating ***non-public*** drinking water systems or receiving water from municipalities should contact the PHC for residual monitoring guidance.

7.3.3 Redundancy of Disinfection Equipment: For surface water systems, replacement disinfection equipment must be available and in service within four hours of problem identification. For groundwater systems, replacement of disinfection equipment varies based off the results of the most frequent sanitary survey or public health assessment.

7.3.4 Special Monitoring - Bacteriological Sampling: The chlorine residual must be measured at the bacteriological sample site each time a bacteriological sample is collected. The monitoring results must be recorded on the operator's log and the laboratory form.

8.0 OPERATOR REQUIREMENTS

- 8.1 All parks that operate public drinking water systems will have certified operators as required by the primacy agency. If the park chooses to contract to have an operator, that operator must adhere to NPS policy and the Primacy Agency. All disinfected non-public water systems require the lowest level of certification available through the State, although a higher level of certification is acceptable.
- 8.2 The park must designate in writing, backup operators who have adequate training and skills to properly operate the system when the primary operator is not available. Equivalent backup operator certification and training is recommended and may be required by some Primacy Agencies.
- 8.3 NPS Unit Managers will develop training plans and assure that operators receive any required and/or appropriate training. The NPS OPH Field Services Branch, in partnership with NPS Facilities Management, provides training on an annual basis or as funding permits.
- 8.4 NPS Unit Managers and/or NPS operators shall notify the regional PHC of any scheduled visits from the primacy agency to inspect NPS owned facilities and provide their contact information, along with the scheduled date and time of the visit from other authorities regulating public drinking water.

9.0 CROSS CONNECTION CONTROL

The Safe Drinking Water Act and International Plumbing Code require all public water systems to maintain an active cross connection control program to protect the health of the public. All cross connections must be protected to prevent contamination of drinking water systems. This is the responsibility of the owner/operator of the system, as well as the operator of each facility connected to the water system.

Each park must have a documented cross connection control program on file for review by the Primacy Agency and/or the PHC. An example policy for cross-connection and backflow control is available from your PHC consultant. For proper backflow prevention in your safe drinking water systems, please refer to Table 3.0 below.

TABLE 3.0 SUMMARY OF BACKFLOW PREVENTION ASSEMBLY APPLICATIONS*

Backflow Assembly	Non-Health Hazard (Pollutant)		Health Hazard (Contaminant)		Sewage	
	Backsiphonage	Backpressure	Backsiphonage	Backpressure	Backsiphonage	Backpressure
AIR GAP (AG)	X	X	X	X	X	X
Reduced Pressure Backflow Assembly (RPBA)	X	X	X	X		
Double Check Valve Assembly (DCVA)	X	X				
Pressure Vacuum Breaker(PVB)	X		X			
Air Vacuum Breaker (AVB)	X		X			

* Adapted from Manual of Cross-Connection Control²⁹, Univ.Southern CA Foundation for Cross-Connection Control and Hydraulic Research

a. Backflow Terminology

Cross-connection: Any unprotected actual or potential connection between a public water system and any other source through which it is possible to introduce any substance other than the intended potable water.

Non-Health Hazard: An impairment of the quality of water to a degree which does not create a hazard to the public but does adversely affect the aesthetic quality of water for domestic use.

Health Hazard: An impairment of the quality of the water, which creates an actual hazard to the public through poisoning or the spread of disease.

Backsiphonage: Backflow due to a reduction in system pressure which causes negative pressure to exist in the water system.

Backpressure: A rise in pressure in the downstream piping system above the supply pressure which would cause a reversal of the normal direction of flow.

Indirect Connection: A cross-connection which is subject to backsiphonage only.

Direct Connection: A cross-connection which is subject to both backsiphonage and backpressure.

²⁹ <http://www.manassacity.org/DocumentCenter/Home/View/2105> Accessed on 6/15/2017

10.0 POTABLE WATER HAULING

In the absence of State or local standards, the following standards shall apply to NPS water hauling operations, whether conducted by the Park or a private contractor:

- 10.1 Water for all NPS water hauling operations, whether conducted by the park or a private contractor, will be obtained from an approved water source that meets the requirements of the Safe Drinking Water Act (SDWA and NPS requirements described in this document).
- 10.2 Containers must be: (1) constructed of non-toxic materials; (2) be non-porous; (3) have never been used for storing or hauling anything but potable water; and (4) be labeled "potable water only". NSF approved containers are recommended.
- 10.3 Before the container is filled, sufficient chlorine shall be added in the water hauled to achieve a free chlorine residual of 1.0 ppm. A free chlorine residual sample shall be taken, and recorded at each location where water is added to a different cistern.
- 10.4 All water haulers will maintain the following records and such records shall be made available to the NPS representatives and NPS PHC upon request:
 1. When and where each load was picked up.
 2. When and where each load was delivered.
 3. A record of all chlorine residual tests.
 4. A record of types and quantities of any chemicals added to the load. (Sodium hypochlorite, etc.)
 5. A maintenance record for the tank and equipment showing method and frequency of cleaning.
 6. Any remarks the hauler feels are pertinent.
- 10.5 The container must be flushed each time water is hauled if it has not been used for more than one day. Water hauling truck shall provide any procedures for hauling to the park and PHC upon request. Primary Agency protocols must be followed.
- 10.6 At no time during the water filling operation shall a potential for backflow exist.
- 10.7 Hoses used to fill and empty tanks shall be properly identified, and used only for potable water. The ends of the hoses shall be capped, when not in use, and the caps shall be attached to the hoses.
- 10.8 All containers, tanks, hoses, fittings, and other equipment used to store, haul, or transfer potable water must be constructed of approved food grade materials or coated with approved food grade coatings.
- 10.9 Hoses shall be stored in such a manner that they are not subject to contamination from surface run-off, birds, rodents, and other sources of contamination.
- 10.10 All valves and fire hydrants (assure hydrant water source meets SDWA and RM83A-1 for nonpublic systems) shall be flushed before the connection of any hoses.

- 10.11 No bacteriological testing will be required on water hauling tanks when source water systems and receiving water systems have a bacteriological monitoring program in place.

11.0 REPORTS AND RECORDS RETENTION

NPS Unit Managers will assure that required records are maintained in permanent files for periodic review by the Public Health Consultant (PHC) or Primacy Agency representatives, and that reports are submitted on a timely basis, as requested by the PHC and/or the Primacy Agency. The results for any testing that is mandated in this policy should be provided to the PHC if not reviewed during the onsite visit. Please consult your PHC for preferred reporting formats, and submission tools.

The park has the responsibility to maintain official records, to have them available in an organized manner for the review and inspection of various regulatory entities, and for periodic review by the PHC. Records must be retained as follows or as required by the Primacy Agency:

<u>RECORDS</u>	<u>RETENTION PERIOD</u>
Chemical Analyses	Indefinitely
Bacteriological Analyses	Five years
Turbidity Measurements	Five years
Public Notices, Administrative Orders, Variances and Exemptions	Five years
Sanitary Surveys	Ten years
Operating Records--includes water usage, water production, chemical usage, chlorine residuals, etc.	Five years
System History - includes well logs test pump data, system modifications, as-builts, operator designations, significant events	Indefinitely
Lead and Copper Rule Records	Twelve years

12.0 WATER CONSERVATION

NPS Management Policies (Chapter 9.1.5.1) requires that the NPS design, construct, manage, and maintain water supply systems in a fashion that promotes conservation. Conservation measures which should be considered include: metering, leak detection and correction, automatic irrigation systems, low-use water fixtures, low flush toilets, and programs which discourage wasteful use of water.

Water Reuse: NPS approves wastewater to be treated for non-potable reuse only, and does not allow the reuse for potable uses. Refer to RM83A-2 for wastewater reuse policy.

13.0 PAYMENT OF FEES FOR SERVICE

Many primacy agencies charge fees for services. Fees are charged for plan review, construction and operating permits, and for sanitary surveys conducted by the primacy agency. The Safe Drinking Water Act placed federally owned and operated systems under the jurisdiction of Primacy Agencies. Parks may be required to pay subject fees. Payment of fees should be included in annual budget and contracting plans, as appropriate.

14.0 PUBLIC NOTIFICATION

All parks will comply with the public notification requirements of the Safe Drinking Water Act³⁰ and the Primacy Agency. The public notification requirements of the Safe Drinking Water Act require water systems to notify persons relative to a three tier system. Tier 1 is an immediate notice, Tier 2 is notice as soon as possible, and Tier 3 is an annual notice (examples³¹).

15.0 PLAN REVIEW AND APPROVAL

When drinking water system modifications or new construction is proposed, parks will contact the Primacy Agency to determine if plans and specifications should be submitted for approval. A copy of the plans and specifications will be provided to the PHC for review. If no regulatory guidance is available for public or non-public systems, American Water Works Association (AWWA) Standards and/or Recommended Standards for Water Works (Ten States Standards) should be followed. The PHC may be contacted for technical review and guidance on public health related issues (e.g. treatment, backflow prevention, disinfection, etc.). The NPS Denver Service Center has accepted codes and standards³².

16.0 OPERATION AND MAINTENANCE PLAN

All parks operating drinking water systems will have a written Operations and Maintenance Plan and an Emergency Response Plan. A copy shall be provided to the PHC for review upon completion. Operation and Maintenance Plan and Emergency Response Plan examples can be found on the NPS OPH website³³.

17.0 POTABLE WATER FOR BACKCOUNTRY OPERATIONS

See RM83A-5 for further details.

18.0 CONSUMER CONFIDENCE REPORTS

The Safe Drinking Water Amendments of 1996 require that Public Community Water Systems provide all consumers with a yearly Consumer Confidence Report which outlines all bacteriological, physical, and chemical monitoring results and Maximum Contaminant Levels for the preceding year.

19.0 DRINKING WATER SECURITY AND EMERGENCY PREPAREDNESS

Security and emergency response planning have always been a critical element of managing a drinking water system. Threat categories include physical destruction, bioterrorism/chemical contamination, cyber-attacks, and personnel disruption. Potential threats can come from vandals, disgruntled employees, insider sabotage, activist groups, or state-sponsored terrorist groups.

The Park should determine if there are any state mandated requirements for providing drinking water security assessments.

If there are no state requirements, the park can utilize the following list to protect the water supply from contamination and other harm:

³⁰ <http://water.epa.gov/lawsregs/rulesregs/sdwa/publicnotification/basicinformation.cfm> Accessed on 6/15/2017

³¹ <https://www.epa.gov/region8-waterops/reporting-forms-and-instructions-public-notification#> Accessed on 6/15/2017

³² <http://www.nps.gov/dscw/ds-civil-environmental.htm> Accessed on 6/15/2017

³³ http://www.nps.gov/public_health/eh/dw.htm Accessed on 6/15/2017

1. Prepare or update an emergency response plan. Ensure all level of involvement and that all staff receive training on the plan.
2. Post updated emergency 24-hour numbers in highly visible areas (pumphouses, vehicles, offices) and give them to key personnel and local response officials.
3. Get to know the Park Law enforcement staff and/or the local law enforcement personnel. Ask them to add your facilities to their routine patrols. Practice emergency response procedures with the rangers, emergency response and public health staff.
4. Fence and lock your drinking water facilities and vulnerable areas (e.g. wellhead, hydrants, manholes, pumphouses, and storage tanks).
5. Lock all entry gates and doors and set any alarms to indicate illegal entry. Do not leave keys in equipment or vehicles at any time.
6. Install good lighting around your pumphouse, treatment facility and parking lot.
7. Identify existing and alternate water supplies and maximize use of backflow prevention devices and minimize interconnections.
8. Use your Source Water Assessment information to lessen any threat posed by potential sources of contamination.
9. Lock monitoring wells to prevent vandals or terrorists from pouring contaminants directly into ground water near your source. Prevent pouring or siphoning contaminates through vent pipes by moving them inside the pumphouse or treatment plant. If that is not possible, fence or screen them.
10. In the event of an emergency, first call “911”, then activate your emergency response plan.

A self-assessment vulnerability checklist is available on the NPS OPH website to help parks identify potential vulnerability concerns for their systems. An in-depth vulnerability profile assessment can be found in the Association of State Drinking Water Administrators, National Rural Water Association “Security Vulnerability Self-Assessment Guide for Small Drinking Water Systems” document that is provided on the NPS OPH³⁴ website, along with other drinking water security references.

20.0 PROTECT SOURCES OF DRINKING WATER³⁵

There are many factors that impact drinking water sources. With global warming concerns drinking water facilities and sources need to assure that they are climate ready. The vulnerability assessment mentioned above is a tool that helps address some new emerging concerns. The following six steps provide water operators, park managers, and PHCs with information to help protect drinking water sources.

1. Delineate the source water protection area (SWPA).
2. Inventory known and potential sources of contamination.
3. Determine the susceptibility of the PWS to contaminant sources or activities within the SWPA.
4. Notify the public about the threats identified in the contaminant source inventory and what they mean for the PWS.
5. Implement management measures to prevent, reduce, or eliminate risks to your drinking water supply.
6. Develop contingency planning strategies that address water supply

³⁴ http://www.nps.gov/public_health/eh/dw.htm Accessed on 6/15/2017

³⁵ <http://www.epa.gov/sourcewaterprotection>. Accessed on 6/15/2017

contamination or service interruption emergencies.

20.1 Disposition of Unused Wells:

Water wells that are in use shall be referred to as active wells, wells that are not in use (such as those that are being used as monitoring wells or that the Park wants to maintain for water rights) shall be referred to as an inactive well and maintained in accordance with the appropriate State standards. If the Park does not use or maintain an inactive well, the well shall be referred to as an abandoned well and destroyed in accordance with State standards. During the time the well is inactive the sanitary seal must be properly maintained to prevent any contamination of the aquifer.

21.0 NATIONAL PRIMARY DRINKING WATER STANDARDS³⁶

According to the EPA website “Under the Safe Drinking Water Act (SDWA), EPA sets legal limits on the levels of certain contaminants in drinking water. The legal limits reflect both the level that protects human health and the level that water systems can achieve using the best available technology. Besides prescribing these legal limits, EPA rules set water-testing schedules and methods that water systems must follow. The rules also list acceptable techniques for treating contaminated water. SDWA gives individual states the opportunity to set and enforce their own drinking water standards, if the standards are at least as strong as EPA's national standards. Most states and territories directly oversee the water systems within their borders.” The National Park Service Office of Public Health thereby follows state and local standards for public water systems and similar standards for non-public systems, if the state does not address those systems.

Information regarding chemical contaminants are referenced in the National Primary Drinking Water Regulations (NPDWRs or primary standards) and are legally enforceable standards that apply to public water systems. Primary standards protect drinking water quality by limiting the levels of specific chemical contaminants that can adversely affect public health and are known or anticipated to occur in public water systems. The link below divides these contaminants into Microorganisms, Disinfection Byproducts, Disinfectants, Inorganic Chemicals, Organic Chemicals, and Radionuclides.

22.0 NATIONAL SECONDARY DRINKING WATER REGULATIONS

National Secondary Drinking Water Regulations (NSDWRs or secondary standards) are non-enforceable guidelines regulating contaminants that may cause cosmetic effects (such as skin or tooth discoloration) or aesthetic effects (such as taste, odor, or color) in drinking water. EPA recommends secondary standards to water systems, but does not require systems to comply. However, states may choose to adopt them as enforceable standards.

- National Secondary Drinking Water Regulations - The complete regulations regarding these contaminants, known as secondary contaminants, is available from the Code of Federal Regulations Web Site³⁷ and appendix 3.

³⁶ <http://water.epa.gov/lawsregs/rulesregs/sdwa/currentregulations.cfm> Accessed on 6/15/2017

³⁷ http://www.access.gpo.gov/nara/cfr/waisidx_02/40cfr143_02.html Accessed 6/15/2017

- For more information and a list of secondary drinking water regulations see the EPA website³⁸.

TABLE 4.0 GENERAL MINERALS*

Contaminant	(mg/L)
Sodium	200 or less is preferred. 20 is considered the level for concern for people with the need to control intake.
Alkalinity	30-500
Calcium	75-200
Hardness	<200
Hydrogen Sulfide	Test at source only when necessary
Magnesium	50-150
Phosphate	<0.2
Potassium	For corrosion control
Specific Conductance	For corrosion control
Temperature (At Source)	For corrosion control

- Values listed are for information only. No limits are established.

³⁸ <https://www.epa.gov/dwstandardsregulations/secondary-drinking-water-standards-guidance-nuisance-chemicals>
 Accessed 6/15/2017

TABLE 5.0 SUMMARY OF PARAMETERS AND TESTING REGIME

CHEMICAL ANALYSIS									
SYSTEM TYPE	SOURCE TYPE	BACTERIOLOGICAL ANALYSIS	CHLORINE RESIDUAL ANALYSIS	TURBIDITY (SURFACE WATER ONLY)	PRIMARY INORGANIC	SECONDARY CONTAMINANTS/GENERAL MINERAL		ORGANICS	
								PESTICIDE/HERBICIDE	VOLATILE
COMMUNITY WATER SYSTEM (CWS)	GW		ONE/DAY More maybe required by GWR (see section 7.1.8)	POPULATION <=500 ONE/DAY POPULATION >500 CONTINUOUS OR EVERY 4 HOURS	EVERY THREE YEARS Annual Nitrates	EVERY THREE YEARS	ANNUAL UNLESS STATE WAIVER ISSUED	EVERY THREE YEARS UNLESS STATE WAIVER ISSUED	Quarterly by the end of 2007, then every 3,6, or 9 years depending on results
					ANNUALLY Annual Nitrates	ANNUALLY			
NON-TRANSIENT NON-COMMUNITY (NTNC)	SW	ONE/MONTH OR PER TABLE 1	SEE TABLE PAGE 15						EVERY NINE YEARS PWS <=3,300 PEOPLE AND MEETS MONITORING WAIVER IN 40 CFR 141.86(g) POPULATION <=500 ONE/YEAR POPULATION >500 ONE/QUARTER POPULATION >10,000 FOUR/QUARTER
Transient Non-Community Water System (TNC)	GW		ONE/DAY		EVERY NINE YEARS Annual Nitrates	EVERY NINE YEARS	ONE TIME AND AS REQUIRED BY PHC	ONE TIME AS REQUIRED BY PHC	AS REQUIRED BY PHC
	SW		SEE TABLE PAGE 15				ONE TIME AND AS REQUIRED BY PHC	AS REQUIRED BY PHC	AS REQUIRED BY PHC
NON-PUBLIC (NP)	GW	ONE/MONTH	THREE/ WEEK or AS REQUIRED BY PHC	DETERMINE ON INDIVIDUAL BASIS	EVERY NINE YEARS Annual Nitrates	EVERY NINE YEARS	ONE TIME AND AS REQUIRED BY PHC	SPECIAL SITUATIONS AS REQUIRED BY PHC	AS REQUIRED BY PHC
	SW								

GLOSSARY OF COMMONLY USED TERMS

Item	Definition
Aquifer	A natural underground layer, often of sand or gravel, which contains water.
Bioterrorism	An agent defined as organisms (including bacteria, viruses, and protozoa) and biotoxins that may be found in natural environmental surroundings or may be weaponized in such a way to increase their virulence and/or increase their resistance to vaccination or antibiotic treatment.
CT-value (Concentration X Time)	The product of "residual disinfection concentration" in mg/L determined before or at the first customer, and the corresponding "disinfection contact time" in minutes.
Cryptosporidium	Coccidian protozoan shed in the feces of man and animals in the form of oocysts, which can survive under adverse environmental conditions for long periods of time. Responsible for a severe gastrointestinal disease called cryptosporidiosis.
<i>Escherichia coli</i> (<i>E.coli</i>)	One of the species of bacteria in the fecal coliform group. It is found in large numbers in the gastrointestinal tract and feces of warm-blooded animals and man. Its presence is considered indicative of fresh fecal contamination, and it is used as an indicator organism for the presence of less easily detected pathogenic bacteria.
Fecal Coliforms	Aerobic and facultative, gram-negative, non-spore-forming rod-shaped bacteria capable of growth at 44.5 degrees C., and associated with fecal matter of warm-blooded animals.
<i>Giardia lamblia</i>	Flagellate protozoan shed in the feces of man and animals, usually in the cyst stage, and responsible for a severe gastrointestinal disease called giardiasis.
Ground water under the direct influence of surface water (GWUDI)	Water beneath the surface of the ground with (1) significant occurrence of insects or other macroorganisms, algae, or large-diameter pathogens such as <i>Giardia lamblia</i> ; or (2) significant and relatively rapid shifts in water characteristics such as turbidity, temperature, conductivity, or pH which closely correlate to climatological or surface water conditions.

Item	Definition
Heterotrophic Plate Count (HPC)	Number of colonies of heterotrophic bacteria grown on selected solid media at a given temperature and incubation period, usually expressed in number of colony forming units per milliliter of sample (CFU/ml).
Heterotrophic microorganisms	Bacteria and other microorganisms that utilize organic matter synthesized by other organisms for energy and growth.
Inorganic Chemicals (IOC)	Chemical substances of mineral origin not having carbon in their molecular structure.
Maximum Contaminant Levels (MCL)	The highest permissible concentration of a substance allowed in drinking water, as established by EPA.
Maximum Contaminant Level Goals (MCLG)	The highest permissible concentration of a substance allowed in drinking water at which no known or anticipated health effects will occur. They are health goals and are not enforceable.
Minimum Detection Limit (MDL)	The lowest achievable concentration of a contaminant that can be measured under ideal laboratory conditions. A more technical definition is the minimum concentration of a substance that can be identified, measured, and reported with 99% confidence that the concentration of the substance being measured is greater than zero.
Non-Potable	Water that may contain objectionable pollution, contamination, minerals, or infective agents and is considered unsafe and/or unpalatable for drinking. If water is non-potable, a clear statement that the water is non-potable and should not be used for drinking, food or beverage preparation, hand washing, teeth brushing, or any other consumptive use should be stated.
Pathogen	A variety of bacteria, protozoan parasites, and viruses
Potable Water	Water that meets the standards for drinking purposes of the State or local authority having jurisdiction, or water that meets quality standards prescribed by the U.S. Environmental Protection Agency.
Potable Water Hauling:	The transportation of potable water as a primary, a supplemental, or an emergency source in containers greater than 50 gallons.
Primacy Agency	Generally a State agency authorized by the EPA to

Item	Definition
	administer provisions of Safe Drinking Water Act. (Note: Not all States have requested this authorization. If authorization is not requested, EPA is the Primacy Agency).
Public Health Assessment	A combination of on-site sanitary surveys of safe drinking water systems, wastewater systems, disease surveillance, food safety inspections, recreational water surveys, public health in backcountry operations, and promoting health in park units.
Radionuclide	A material with an unstable atomic nucleus, which spontaneously decays or disintegrates, producing radiation.
Sanitary Survey	An onsite review of the water source, facilities, equipment, operation and maintenance of a public water system for the purpose of evaluating the adequacy of such source, facilities, equipment, operation and maintenance for producing and distributing safe drinking water.
Source Water	Surface water (streams, rivers, and lakes) or ground water (aquifers) can serve as sources of drinking water, referred to as source water . Source water provides water for public drinking water supplies and private water wells.
Synthetic Organic Chemicals (SOC)	Man-made organic chemicals, many of which have been detected in drinking water. This group includes the VOCs.
Trihalomethanes (THM)	A group of volatile organic compounds formed when chlorine reacts with naturally occurring humic substances. Individual Compounds are not regulated under THM's.
Total Coliforms (Pathogens)	A group of bacteria predominantly inhabiting the intestines of man or animals but occasionally found elsewhere. Presence in water is used as indication of possible pathogen contamination. Pathogens include various types of bacteria, viruses, protozoan parasites, and other organisms.
Total Trihalomethanes (TTHM)	A term used to designate the total concentration of chloroform, bromoform, dibromochloromethane, and

Item	Definition
	bromodichloromethane in the National Primary Drinking Water Regulations. Combination of THM compounds is regulated.
Volatile Organic Compounds (VOC)	Lightweight, man-made organic compounds that vaporize, or evaporate, easily.
Waiver	The relinquishing of the requirements to perform certain monitoring and/or sampling procedures.

APPENDIX 2.0 FREQUENTLY USED ABBREVIATIONS AND ACRONYMS

AWWA:	American Water Works Association
CFR:	Code of Federal Regulations
CT:	value (Concentration X Time)
CWS:	Community Water System
DBP:	Disinfection Byproduct
DBPR:	Disinfection Byproduct Rule
GW:	Ground Water
GWR:	Ground Water Rule
GWUDI:	Ground water under the direct influence of surface water
HAA5:	Haloacetic acids
HPC:	Heterotrophic Plate Count
IOC:	Inorganic Chemicals
MCL:	Maximum Contaminant Level
MCLG:	Maximum Contaminant Level Goals
MDL:	Minimum Detection Limit
NPS:	National Park Service
NSF:	formerly known as National Science Foundation and NSF International as of 1990
NTNC:	Non-Transient Non-Community
NTU:	Nephelometric Turbidity Unit
NP:	Non-Public
OPH:	Office of Public Health
PHC:	public health consultant
PWS:	Public Water System
PWS:	Public Water System
RM:	Reference Manual
RTCR:	Revised Total Coliform Rule
SDWA:	Safe Drinking Water Act
SOC:	Synthetic Organic Chemicals
SW:	Surface Water
SWPA:	Source Water Protection Area
TC+:	Total Coliform positive
THM:	Trihalomethanes
TNC:	Transient Non-Community
TTHM:	Total Trihalomethanes
USC:	United States Code
USEPA:	United States Environmental Protection Agency

VOC: Volatile Organic Compounds
WQP: Water quality parameter

APPENDIX 3.0 TABLE OF SECONDARY DRINKING WATER STANDARDS

Contaminant	Secondary MCL	Noticeable Effects above the Secondary MCL
Aluminum	0.05 to 0.2 mg/L*	colored water
Chloride	250 mg/L	salty taste
Color	15 color units	visible tint
Copper	1.0 mg/L	metallic taste; blue-green staining
Corrosivity	Non-corrosive	metallic taste; corroded pipes/ fixtures staining
Fluoride	2.0 mg/L	tooth discoloration
Foaming agents	0.5 mg/L	frothy, cloudy; bitter taste; odor
Iron	0.3 mg/L	rusty color; sediment; metallic taste; reddish or orange staining
Manganese	0.05 mg/L	black to brown color; black staining; bitter metallic taste
Odor	3 TON (threshold odor number)	"rotten-egg", musty or chemical smell
pH	6.5 - 8.5	low pH: bitter metallic taste; corrosion
		high pH: slippery feel; soda taste; deposits
Silver	0.1 mg/L	skin discoloration; graying of the white part of the eye
Sulfate	250 mg/L	salty taste
Total Dissolved Solids (TDS)	500 mg/L	hardness; deposits; colored water; staining; salty taste
Zinc	5 mg/L	metallic taste

*mg/L is milligrams of substance per liter of water.

APPENDIX 4.0 NPS OFFICE OF PUBLIC HEALTH DRINKING WATER MONITORING WAIVER APPLICATION

The National Park Service Office of Public Health understands that there may be specific situations where a waiver for water sampling may be necessary. However, public health will not be compromised to issue a waiver and the application below is necessary to help assure proper public health can be maintained within the water system if a waiver is permitted. In order to be considered for a monitoring waiver, the complete application must be submitted at least sixty (60) days prior to system startup (seasonal only) or January 1 of the request year of implementation. Filling out this form does not automatically grant a waiver.

Monitoring Waiver Form

Date:		Park Name:		Name of Person Filling out Application and Title:			
Address:		City/State/ZIP:		County:		Phone number	
Waiver Requested for the Following: Use (X) to select and explain if other.		Bacteriological		Disinfection by-Products		Radionuclides, Other	

NAME OF SYSTEM/PWSID	2	3	4	5	6
Please write the system name and public water system ID (PWSID) (if applicable) in the spaces provided to the right. → Have all NPS Reference Manual 83A testing requirements been met? NA means not applicable. Note: The PHC may not issue waivers for items that are regulated by Primacy Agency (see RM83A for questions). *No does not mean omit the possibility of a waiver.					
	1. System type per RM83A (CWS, NTNC, TNC, NP)				
	2. Primary Inorganic chemicals (IOCs), Secondary Contaminants, & General Minerals (Yes, No*, or NA)				
	3. Nitrate and Nitrites (Yes, No*, or NA)				
	4. Organics: Pesticides and Herbicides (Yes, No*, or NA)				
	5. Organics: Volatile Organic Chemicals (VOCs) (Yes, No*, or NA)				
	6. Radionuclides (Yes, No*, or NA)				
	7. Lead and Copper (Yes, No*, or NA)				
	8. Disinfection by-products (Yes, No*, or NA)				
	9. Bacteriological (Yes, No*, or NA)				
	10. Chlorine Residuals (Yes, No*, or NA)				
11. Type of Source (Check or "X" under the appropriate source)					
a. Ground Water (including springs)					
b. Surface Water (including ground water sources under the influence of surface water)					
12. Land use and system characteristics (Write in appropriate number under each source)					
a. Land use within 2-mile radius of source (round to the nearest 5% and list all known land uses and total up to 100%): 1. Residential 2. Agricultural—livestock 3. Agricultural—crop land 4. Industrial/Manufacturing 5. Undeveloped 6. Mining 7. Other (describe on separate paper and attach					

b. System has a source water/drinking water protection plan. Yes or No*								
c. Is the source influenced by geothermal waters?								
d. Have there been any changes to the system's configuration or pumping rates within the past year? If yes, please submit a description of the changes along with the application.								
e. Are there any other land use practices (farming, industrial, mining) within a 10 mile radius that would create public health concerns for the water source? If yes, approximate distances and list uses								
13. Well characteristics (Only fill out this portion if the source is a well)								
a. Please indicate the depth of the well in feet at the right (Depth to end of casing, 1 st screen or perforation in casing) Check well log or source water assessment. Leave blank or write "NA" if not known.								
c. Construction date and description of well.								
b. Approximate distance in miles to nearest active/inactive dump or landfill. NA if not known or >5 mi								
d. Are there any of the following potential sources of contamination within 300 feet of the wellhead (write Yes or No) If yes, please attach a description and an estimated location of the facility to the wellhead. Service stations, dry cleaners, small or large machinery repair shops, electronics repair shops, other light/heavy industry, or wastewater systems.								
e. Did the most recent public health assessment indicate that the well meets minimum construction standards for flood protection? (write Yes or No or UNK for unknown)								
f. Indicate by number the best description of the soil type around the well from about 0-20 feet below the surface: 1. Topsoil rich in humus 2. Topsoil with heavy clay 3. Sand or silt 4. Fractured rock or gravel 5. Solid rock 6. Other 7. Unknown								
14. Surface water characteristics (Only fill out this portion if the source is surface water)								
a. Has the watershed been inspected in the past year for signs of contamination? Yes or No .								

The information provided is true and accurate to the best of my knowledge:

NPS Office of Public Health Regional Consultant:



NPS Office of Public Health Director:

Signature of Authorized Agent and Title

Date

Signature and Title

Date

Signature and Title

Date

Determining vulnerability for possible monitoring waiver

For Source vulnerability, we try to identify potential contamination from activities that may occur in the area from which the source draws its water. Two factors influence used to access vulnerability are:

1. **Physical susceptibility to contaminant infiltration:** Physical susceptibility is determined by examining conditions that affect the movement of contaminants from the land surface into a water supply. This includes the depth of the well, its construction, the geology of the area, the pumping rate, the source(s) of groundwater recharge, and the aquifer material.

2. **Contaminant exposure:** The risk of contaminant exposure is based on what contaminants, if any, are used in the water supply area.

NOTE: This waiver will be accompanied by a memo that outlines what the park will do to still ensure potable water is provided. As a reminder, this waiver is valid for one year and can be canceled at any time if a public health issue is identified and confirmed throughout the waiver year.

It can be difficult to predict groundwater pollution from surface exposure accurately because each contaminant type may act differently in the environment. Therefore, physical susceptibility helps to determine vulnerability. If physical susceptibility data is incomplete, or contaminant use is highly unlikely, we base the vulnerability assessment on risk of exposure to contaminants. If there are questions or comments, please contact your regional public health specialist (https://www.nps.gov/public_health/)

APPENDIX 5.0 DRINKING WATER EXAMPLE SITE SAMPLING PLAN (FROM EPA REGION 8)

Sample Siting Plan Instructions

22.1 **Why do I need to develop a Revised Total Coliform Rule (RTCR) sample siting plan?**

The purpose of this plan is to specify where in the distribution system routine bacteriological samples will be collected to ensure they are representative of the water quality in your system. The original siting plan should be kept on-site for use by sampling personnel. Beginning April 1, 2016, under the RTCR, every water system will be required to collect bacteriological samples monthly. Prior to that date, your system will remain on the schedule as indicated in the annual Monitoring and Reporting Requirements list that EPA sends to your system.

The RTCR requires EPA to collect and review every public water system's RTCR sample siting plan prior to April 1, 2016, so please review, revise and submit your plan before that date. You will need to submit:

1. A list of sampling locations (a blank chart can be found in **Addendum C**):
 - You will need to indicate the sites for routine and repeat bacteriological monitoring in your distribution system; and
 - Any source water sampling sites if subject to the Ground Water Rule.
2. A map of the distribution system showing locations of your sample sites as described in the sample siting plan. You can use a diagram, distribution system map, aerial photo, etc. Clearly indicate if there are multiple distribution systems and if those distribution systems are connected to each other.

22.2 **Where to submit your plan:**

Send a copy of your RTCR sample siting plan via mail, e-mail, or fax to:

The Privacy Agency
and/or
The regional National Park Service Office of Public Health Public Health Consultant

22.3 **How to choose a laboratory:**

Not all laboratories have been approved and certified to analyze for all the required drinking water contaminants. You must send your samples to a laboratory that is certified for the specific samples that you are submitting. For a list of certified labs, visit the EPA Region 8 website: <http://www2.epa.gov/region8-waterops> and click the certified lab link.

General Requirements

22.4 *Process for selecting sample sites and rotation:*

Review the layout of your distribution system and choose RTRC sample sites that will represent each area of the distribution system if sampled on a monthly rotating basis throughout an entire year or open season. Be sure to take into account non-permanent sources (seasonal or interim). These sources need to be represented within your siting plan. If your water system has multiple (completely separated) distribution systems, you must select sample sites within each separate distribution system. Samples can be rotated among the different distribution systems in different months, or you can collect samples in each distribution system each month, even if that means collecting more samples than are required. See Addendum B, example 3.

The siting plan may need to be updated periodically to account for system changes (such as population changes, new sources, or change in treatment, etc). The plan should be reviewed annually and during your sanitary survey to incorporate any changes. Submit any revisions to the EPA regional office as soon as they occur.

22.5 *Routine Monitoring Requirement*

Unless otherwise specified in your monitoring and reporting requirements, most systems will be placed on the monitoring schedule seen in **Addendum A** of this document.

Use the chart below as an example for formatting the sample siting plan. The following must be included in the plan:

- Routine sampling location(s);
- Repeat sampling locations (only necessary if your routine sample is TC+);
- Ground Water Rule (GWR) source sample locations for systems using ground water sources. (GWR source samples are only required if your routine sample is TC+. You must sample every ground water source operating at the time of the TC+ routine sample).

22.6 *Repeat Monitoring*

After April 1, 2016 under the RTRC, systems must collect no fewer than three repeat samples for each TC+ routine sample. This requirement will apply to all systems.

- All repeat samples need to be taken within 24 hours of notification of a TC+ routine sample. If you cannot make this timeframe, be sure to contact EPA Region 8 as soon as possible;
- One repeat sample is required to be taken from the same tap as the original TC+ sample;
- One repeat sample should be taken within five taps upstream of the original TC+ sample, and one should be taken within five taps downstream;

Take note of your routine sample locations. Can you take proper repeats based on your current choices of routine sampling sites? If not, you will need to reevaluate your routine sites and choose another location. If you do not have a distribution system, upstream and downstream locations may not exist. In this case, you would need to collect multiple samples at the original location to count as repeat samples.

Follow this same format but expand as necessary if your system would need to take more than one sample per month. (Seasonal systems only have to sample during their operating season). See **Addendum B** for examples and **Addendum C** for blank pages.

1 sample/month	<u>Routine sample location</u>	<u>Repeat sample locations</u>	<u>GWR sample location (any ground water sources operating at the time of a routine TC+)</u>
January	1)	1) (same as routine location)	
	(within 5 taps upstream)	2)	
	(within 5 taps downstream)	3)	
February	1)	1) (same as routine location)	
		2)	
		3)	
March	1)	1) (same as routine location)	
		2)	
		3)	

E. coli Testing and E. coli MCL Compliance Determination

Any TC+ sample result is automatically analyzed for *E. coli* (EC) by your lab. Any EC+ repeat sample or any total coliform-positive repeat sample following an EC+ routine sample constitutes an *E. coli* MCL violation. All *E. coli* MCL violations are situations that require you to contact EPA Region 8 immediately and distribute public notice including a boil order to your customers.

Disinfectant (chlorine) residual monitoring:

If you chlorinate, your system may need to conduct residual disinfectant monitoring in the distribution system for the Disinfection By-Product Rule and/or the Surface Water Treatment Rule. Review your Monitoring and Reporting Requirements sheet to verify if disinfectant monitoring is necessary. If you need to conduct this monitoring, keep the following in mind:

- The residual disinfectant must be measured at the same time and the same location as each total coliform bacteria sample.
- These measurements must be conducted in the field by a certified operator (or under the direction of the certified operator).
- Residual disinfectant measurements must be written on each total coliform sample slip when it is submitted to the laboratory.
- Ask your laboratory to forward this information to EPA along with the sample results.

Depending on the disinfectant used in the distribution system, ensure you are measuring the proper disinfectant residual:

- ❖ If chlorine is used, the disinfectant residual must be measured as free, total, or combined chlorine
- ❖ If chloramines are used, the disinfectant residual must be measured as total chlorine

Ground water source sampling:

22.8 Triggered Source Water Monitoring Requirement

Your system may need to conduct triggered source water sampling if you use a ground water source and have a TC+ routine sample result. Review your Monitoring and Reporting Requirements sheet to verify if source water monitoring is necessary. If you need to conduct this monitoring, keep the following in mind:

- Within 24-hours of notification that a *routine* Total Coliform Rule distribution system sample is TC+, you must collect a raw water sample from each groundwater source that was in use at that time for every routine TC+ sample (e.g. if you have three routine TC+ samples, you will need to collect three source samples from *each* ground water source). This sample must be analyzed for E. coli.
- You will need to report this data using the “Source Water Sampling-Triggered Source Monitoring Sample Collection and Reporting Form” found on our website at: <http://www2.epa.gov/region8-waterops/wyoming-and-tribal-triggered-groundwater-source-sampling-form>
- If you need further assistance on the Ground Water Rule and the required steps after a routine TC+ sample, please consult: <http://www2.epa.gov/region8-waterops/epa-region-8-drinking-water-unit-tech-tips-follow-unsafe-total-coliform-positive>

If the system received EPA Region 8 approval to use a sampling site that represents more than one groundwater source, please indicate that on the sample collection and monitoring form mentioned above that is submitted with your samples.

22.9 **Addendum A:** *Required number of monthly routine samples under the Revised Total Coliform Rule*

Sampling requirements for all water systems:

Population served/day	Routine Samples/Month	Population served/day	Routine Samples/Month
25-1,000	1	8501-12900	10
1001-2500	2	12901-17200	15
2501-3300	3	17201-21500	20
3301-4100	4	21501-25000	25
4101-4900	5	25001-33000	30
4901-5800	6	33001-41000	40
5801-6700	7	41001-50000	50
6701-7600	8	50001-59000	60
7601-8500	9	59001-70000	70

22.10 **Addendum B:** *Examples of RTCR Sample Siting Plans*

- 1) Your water system is a town serving 550 people year around. One routine TCR sample per month is required. Repeating the same set of three rotating locations every quarter is acceptable. Repeat this pattern to finish the monitoring year.

1 sample/month	<u>Routine sample location</u>	<u>Repeat sample locations</u>	<u>GWR sample location (any sources running at the time of a routine TC+)</u>
January	1) Town Hall	1) (same as routine location)	Well #1
	(within 5 taps upstream)	2) Senior Center	Well #2
	(within 5 taps downstream)	3) Big A Truck Stop	
February	1) 123 Main St	1) (same as routine location)	Well #1
		2) Oskar's Bar	Well #2

			3) Fire Station	
March	1) 456 Center St.		1) (same as routine location)	Well #1
			2) 789 Center St.	Well #2
			3) 1010 Grand Ave	
April	1) Town Hall		1) (same as routine location)	Well #1
			2) Senior Center	Well #2
			3) Big A Truck Stop	
May	1) 123 Main St		1) (same as routine location)	Well #1
			2) Oskar's Bar	Well #2
			3) Fire Station	
June	1) 456 Center St		1) (same as routine location)	Well #1
			2) 789 Center St.	Well #2
			3) 1010 Grand Ave	

2) You have a seasonal campground/RV park open from June to August that serves 50 people per day but has two separate distribution loops (one for the campground and one for the RV section). Your system would only be required to monitor one sample per month. You could rotate the single sample each month between the two distribution loops, but we recommend collecting a sample from each section each month to ensure both loops are functioning properly. If you opt to take a secondary sample in addition to your routine sample, be sure to label that as "special" when submitting to the lab:

1 sample/month	<u>Routine sample location</u>	<u>Repeat sample locations</u>	<u>GWR sample location (any sources running at the time of a routine TC+)</u>
June	1) CG Loop Tent #1	1) (same as routine location)	Resort Well #1
	2) Optional RV loop sample (special)	2) Shower House	
		3) Rec Room	
July	1) RV loop Space #6	1) (same as routine location)	Resort Well #1
	2) Optional CG loop sample (special)	2) RV loop Space #1	
		3) RV loop Space #10	
August	1) CG Loop Tent #1	1) (same as routine location)	Resort Well #1
	2) Optional RV loop sample (special)	2) Shower House	
		3) Rec Room	

3) You have a system that serves 7,000 people per day. A total of eight samples are required per month and should be collected at regular time intervals throughout the month. For example, you might opt to take a set of four samples on the 1st and 3rd weeks of each month.

First week of the month:	Sample 1) Sample 2) Sample 3) Sample 4)	Third week of the month:	Sample 1) Sample 2) Sample 3) Sample 4)	Senior Center Hospital 789 Old Hwy 123 Main St.
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Use the sample siting form to indicate the repeat locations and any required source water sample locations for each sample site.

Addendum C: **Example forms.** Expand the following table as necessary. Feel free to detach and use this example form:

	<u>PWS NAME:</u>	<u>PWS ID #:</u>	<u>GWR sample location (any sources running at the time of a routine TC±)</u>
(1 sample/mo)	<u>Routine sample location</u>	<u>Repeat sample locations</u>	
January	1)	1) (same as routine location)	
		2)	
		3)	
February	1)	1) (same as routine location)	
		2)	
		3)	
March	1)	1) (same as routine location)	
		2)	
		3)	
April	1)	1) (same as routine location)	
		2)	
		3)	
May	1)	1) (same as routine location)	
		2)	
		3)	
June	1)	1) (same as routine location)	
		2)	

			3)		
July	1)		1) (same as routine location)		
			2)		
			3)		
August	1)		1) (same as routine location)		
			2)		
			3)		
September	1)		1) (same as routine location)		
			2)		
			3)		
October	1)		1) (same as routine location)		
			2)		
			3)		
November	1)		1) (same as routine location)		
			2)		
			3)		
December	1)		1) (same as routine location)		
			2)		
			3)		

APPENDIX 6.0 GENERAL TNC WATER SYSTEM SEASONAL START-UP AND SHUT-DOWN

Key Principles and Procedures for Start-Up

- ☐ Review all system manuals and job-hazard-analyses prior to start-up activities.
- ☐ Follow lock-out and tag-out procedures for electrical service to pumping system during start-up and operational periods.
- ☐ Resupply chemicals following MSDS requirements and store properly onsite.
- ✓ ***Prevent damage to system.***
 - ☐ Operate all valves in the system for proper operation and orientation.
 - ☐ Read gauges, meters and panels upon system start-up to ensure correct settings.
 - ☐ Inspect all exposed plumbing joints, taps, fixtures for leaks or incorrect orientation.
 - ☐ Inspect reservoir/tanks and lines for leaks or damage. Check pressure tanks for waterlogging or pressure issues.
 - ☐ Inspect connection of chemical injection pumps.
 - ☐ Check that electrical components are properly grounded.
 - ☐ Check that doors, fencing and locks are working properly.
 - ☐ Inspect housing of facilities for rodent or insect infestation.
- ✓ ***Prevent contamination of water.***
 - ☐ Flush and disinfect lines and reservoir/tanks. For chlorination and dechlorination procedures, please use the references found at: <http://www.vita-d-chlor.com/> or <https://portal.doi.net/nps/PHP/DW/Shared%20Documents/Forms/AllItems.aspx>
 - ☐ Assure disinfectant residuals are at least 0.2 mg/L and less than 4 mg/L (for chlorine), and report results to the public health consultant, state, and others as required.
 - ☐ Conduct two consecutive bacteriological tests (Per NPS RM83A-1).
 - Assure quality is met in all main legs of the distribution sys. see Revised Total Coliform Rule
 - ☐ Test reservoir level controls and automatic systems.
 - ☐ Test backflow prevention devices. Ensure all vacuum breakers are in place.
 - ☐ Replace defective screens and seals around pipe and reservoir/tank openings.
 - ☐ Inspect well protection area (100 feet) for any sources of contamination.
 - ☐ Inspect well cap and vent for any openings allowing debris, precipitation or animal access.
 - ☐ Check secondary containment around backup power system or fuel containers.
- ✓ ***Communicate system status to stakeholders***
 - ☐ Notify concessioners, visitors, partners, park staff and regional public health consultant at least one week prior to system start-up.
 - ☐ Complete necessary forms required by State to indicate system has restarted.
 - ☐ Place orders for water analysis tests as needed.
 - ☐ Record initial meter readings and any other tracked data.
 - ☐ Start-up each system with ample time to allow for bacteria testing results and all of the above areas to be addressed before the system is in service.

References

[Ohio EPA Seasonal Noncommunity Public Water Systems Start-up and Shut-down Checklist Washington State DOH Small Water System Start-up Shut-down Self-Inspection Checklist](#)

Key Principles and Procedures for Shut-Down

- ☐ Review all system manuals and job-hazard-analyses prior to shut-down activities.
- ☐ Follow lock-out and tag-out procedures for lock-out and tag-out electrical service to pumping system during draining and seasonal shut-down.
- ☐ Move stored chemicals following MSDS requirements to proper ventilated storage area.
- ✓ ***Prevent damage to system.***
 - ☐ Drain reservoirs/tanks and lines. Drain and deactivate water appliances, such as hot water tanks, ice makers, etc.
 - ☐ Properly dispose of remaining chemical injection vat contents.
 - ☐ Run fresh water through chemical injection pumps and drain.
 - ☐ Check that electrical components are properly grounded.
 - ☐ Check that doors, fencing and locks are working properly.
 - ☐ Inspect housing of facilities for rodent or insect infestation.
- ✓ ***Prevent contamination of water.***
 - ☐ Replace defective screens and seals around pipe and reservoir/tank openings.
 - ☐ After draining and removing equipment as needed, close all valves and cap all openings to lines, chemical vats and reservoir/tanks.
 - ☐ Do not add anti-freeze to any parts of the system. This include propylene glycol see section 3.2.1 above.
 - ☐ Inspect well protection area (100 feet) for any sources of contamination.
 - ☐ Inspect well cap and vent for any openings allowing debris, precipitation or animal access.
 - ☐ Check secondary containment around backup power system or fuel containers.
- ✓ ***Communicate system status to stakeholders***
 - ☐ Notify concessioners, visitors, partners, park staff and regional public health consultant at least one week prior to system shut down.
 - ☐ Complete necessary forms required by State to indicate system will be shut down.
 - ☐ Ensure final water analysis tests are complete and sent to the public health consultant and others as needed.
 - ☐ Record final meter readings and any other tracked data.

References

[Ohio EPA Seasonal Noncommunity Public Water Systems Start-up and Shut-down Checklist Washington State DOH Small Water System Start-up Shut-down Self-Inspection Checklist](#)

APPENDIX 7.0 TEMPLATE POLICY FOR THE CONTROL OF BACKFLOW AND CROSS-CONNECTIONS

1.0 Section 1. Cross-Connection Control Policy

1.1 Purpose. The purpose of this policy is:

- 1.1.1 To protect the public and non-public water supply of **[Name of Park Here]** from the possibility of contamination or pollution.
- 1.1.2 To promote the elimination or control of existing cross-connections, actual or potential, between the potable water system(s) and non-potable water system(s), plumbing fixtures and industrial piping systems; and,
- 1.1.3 To provide for the maintenance of a continuing program of cross-connection control which will systematically and effectively prevent the contamination or pollution of all potable water systems.

1.2 Responsibility. The **[Park Position - Chief of Maintenance./Facilities Management suggested here]** shall be responsible for the protection of the public and non-public potable water distribution systems from contamination or pollution through the water service connections. If, in the judgement of the **[Park Position - Chief of Maintenance./Facilities Management suggested here]**, Regional Public Health Consultant, or appropriate regulatory officials, an approved backflow prevention assembly is required for the safety of the water system, The **[Park Position - Chief of Maintenance./Facilities Management suggested here]** shall install, or have installed, such an approved backflow prevention assembly.

2.0 Section 2. Definitions.

2.1 Park Chief of Maintenance/Facilities Manager. The **[Park Position - Chief of Maintenance./Facilities Management suggested here]** is in charge of the park maintenance and is invested with the authority and responsibility for the implementation of an effective cross-connection control program and for the provisions of this policy.

2.2 Approved. Accepted by the **[Park Position - Chief of Maintenance./Facilities Management suggested here]** as meeting an applicable specification stated or cited in this ordinance, or as suitable for the proposed use.

2.3 Auxiliary Water Supply. Any water supply on or available to the premises, other than the National Park Service approved potable water system, will be considered an auxiliary water supply. These auxiliary waters may include water from another purveyor's public potable water supply or any natural source(s) such as a well, spring, river, stream, lake, etc., or "used waters" or "industrial Fluids". These waters may be contaminated or polluted or they may be objectionable and constitute an unacceptable water source over which the National Park Service has no sanitary control.

2.4 Backflow. The reversal of the normal flow of water caused by either backpressure or

backsiphonage.

- 2.5 Backpressure.** The flow of water or other liquids, mixtures or substances under pressure into the pipes of a potable water supply system from any source(s) other than the intended source.
- 2.6 Backsiphonage.** The flow of water or other liquids, mixtures or substances into the distribution pipes of a potable water supply from any source(s) other than its intended source caused by the reduction of pressure in the potable water supply system.
- 2.7 Backflow Preventer.** An assembly or means designed to prevent backflow.
- 2.7.1 Air gap.** The unobstructed vertical distance through the free atmosphere between the lowest opening from any pipe or faucet supplying water to a tank, plumbing fixture, or other device and the flood level rim of the vessel. An approved air gap shall be at least double the diameter of the supply pipe, measured vertically, above the overflow rim of the vessel; and in no case less than one inch.
- 2.7.2 Reduced Pressure Principle Assembly.** An assembly of two independently acting approved check valves together with a hydraulically operating, mechanically independent differential pressure relief valve located between the check valves and at the same time below the first check valve. The unit shall include properly located resilient seated test cocks and resilient seated shut-off valves at each end of the assembly. The entire assembly shall meet the design and performance specifications as determined by the Foundation for Cross Connection Control and Hydraulic Research at the University of Southern California (USC), University Park, Los Angeles, CA 90089., for backflow prevention assemblies. The assembly shall operate to maintain the pressure in the zone between the two check valves at an acceptable level less than the pressure on the potable water supply side of the assembly. At cessation of a normal flow the pressure between the two check valves shall be less than the pressure on the public supply side of the assembly. In case of leakage of either of the check valves, the differential relief valve shall operate to maintain the reduced pressure in the zone between the check valves by discharging to the atmosphere. When the inlet pressure is two pounds per square inch or less, the relief valve shall be open to the atmosphere. To be approved, these assemblies must be readily accessible for in-line testing and maintenance and be installed in a location where no part of the assembly will be submerged.
- 2.7.3 Double Check Valve Assembly** An assembly of two independently operating approved check valves with resilient seated shut-off valves on each end of the check valves, plus properly located resilient seated test cocks for the testing of each check valve. The entire assembly shall meet the design and performance specifications as determined by the aforementioned Foundation for Cross Connection Control at USC for backflow prevention assemblies. To be approved, these assemblies must be readily accessible for in-line testing and maintenance.
- 2.8 Contamination.** Means an impairment of the quality of the potable water by sewage, industrial fluids or waste liquids, compounds or other materials to a degree which creates an actual or potential hazard to the public health through poisoning or through the spread of disease.
- 2.9 Cross Connection.** Any physical connection or arrangement of piping or fixtures between two otherwise separate piping systems one of which contains potable water and the other

nonpotable water or industrial fluids of questionable safety, through which, or because of which, backflow may occur into the potable water system. This would include any temporary connections, spools, dummy section of pipe, swivel or change-over devices or sliding multipoint tube.

- 2.10 Cross Connections** - Controlled. A connection between a potable water system and a nonpotable assembly properly installed and maintained so that it will continuously afford the protection commensurate with the degree of hazard.
- 2.11 Cross Connection Control by Containment.** The installation of an approved backflow prevention assembly at the water service connection to any building where it is physically and economically infeasible to find and permanently eliminate or control all actual or potential cross connections within the building owner's water system; or, it shall mean the installation of an approved backflow prevention assembly on the service line leading to and supplying a portion of a building's water system where there are actual or potential cross connections which cannot be effectively eliminated or controlled at the point of the cross connection.
- 2.12 Hazard, Degree of.** The term is derived from an evaluation of the potential risk to public health and the adverse effect of the hazard upon the potable water system.
- 2.12.1 Hazard - Health.** Any condition, device, or practice in the water supply system and its operation which could create, or in the judgement of the **[Park Position - Chief of Maintenance./Facilities Management suggested here]**, Regional Public Health Consultant, or the appropriate area regulatory officials, may create a danger to the health and well-being of the water consumer.
- 2.12.2 Hazard - Plumbing.** A plumbing type cross-connection in a consumer's potable water system that has not been properly protected by an approved air gap or approved backflow prevention assembly.
- 2.12.3 Hazard - Pollution.** An actual or potential threat to the physical properties of the water system or the potability of the water supply which would constitute a nuisance, be aesthetically objectionable, or could cause damage to the system or its appurtenances, but would not be dangerous to health.
- 2.12.4 Hazard - System.** An actual or potential threat of severe damage to the physical properties of the potable water system, or of a pollution or contamination which would have a protracted effect on the quality of the potable water in the system.
- 2.13 Industrial Fluids System.** Any system containing a fluid or solution, which may be chemically, biologically or otherwise contaminated or polluted in a form or concentration such as would constitute a health, system, pollution or plumbing hazard if introduced into an approved water supply. This may include, but not be limited to: pollution or contaminated waters, all types of processed waters and "used waters" originating from the public potable water system which may be deteriorated in sanitary quality; chemicals in fluid form; plating acids and alkalies; circulating cooling waters connected to an open cooling tower and/or cooling towers that are chemically or biologically treated or stabilized with toxic substances; contaminated natural waters such as from wells, springs, streams, rivers, irrigation canals, or

systems, etc; oils gases, gaseous fluids used in industrial or other purposes or for firefighting purposes, recirculating hot water systems, hot water heating systems, solar systems with antifreeze solutions, and charged fire systems.

2.14 Pollution. Means the presence of any foreign substance (organic, inorganic or biological) in the water which tends to degrade the water quality so as to constitute a hazard or impair the usefulness or quality of the water to a degree which does not create an actual hazard to the public health but which does adversely and unreasonably affect such waters for domestic use.

2.15 Water - Potable. Any water which, according to recognized standards, is safe for human consumption.

2.16 Water - Nonpotable. Water which is not safe for human consumption or which is of questionably quality.

3.0 Section 3. Requirements.

3.1 Water System

3.1.1 The water system shall be considered as made up of two parts: the utility system and the building system.

3.1.2 The utility system shall consist of the source facilities and the distribution system; and shall include all those facilities of the water system up to the point where the building system begins.

3.1.3 The source shall include all components of the facilities utilized in the production, treatment, storage, and delivery of water to the distribution system.

3.1.4 The distribution system shall include the network of conduits used for the delivery of water from the source to the building system.

3.1.5 The building system shall include those parts of the facilities beyond the termination of the distribution system which are utilized in conveying delivered domestic water to points of use.

3.2 Policy

3.2.1 No water service connection to any premises shall be installed or maintained by the National Park Service unless the water supply is protected by State laws and regulations and this Cross Connection and Backflow Prevention Policy. Service of water to any premises shall be discontinued by the National Park Service if a backflow prevention assembly required by this Cross Connection and Backflow Prevention Policy is not installed, tested and maintained, or if it is found that a backflow prevention assembly has been removed, by-passed, or if any unprotected cross connection exists on the premises. Service will not be restored until such conditions or defects are corrected.

3.2.2 The building system should be open for inspection at all reasonable times to authorized

representatives of the National Park Service to determine whether cross connections or other structural or sanitary hazards, including violations of these regulations exist. When such a condition becomes known, the **[Park Position - Chief of Maintenance./Facilities Management suggested here]** shall deny or immediately discontinue service to the premises by providing for a physical break in the service line until the building owner has corrected the condition(s) in conformance with the State and Local statutes relating to plumbing and water supplies and the regulations adopted pursuant thereto.

3.2.3 An approved backflow prevention assembly shall also be installed on each service line to a building water system at or near the property line if relevant, or immediately inside the building being served, but , in all cases, before the first branch line leading off the service line whenever the following conditions exist:

3.2.3.1 In the case of premises having an auxiliary water supply which is not or may not be of safe bacteriological or chemical quality and which is not acceptable as an additional source by the **[Park Position - Chief of Maintenance./Facilities Management suggested here]** or Regional Public Health Coordinator, the public water system shall be protected against backflow from the premises by installing an approved backflow prevention assembly in the service line appropriate to the degree of hazard.

3.2.3.2 In the case of premises on which industrial fluids, sewage, or any other objectionable substance is handled in such a fashion as to create an actual or potential hazard to the public water system, the public water system shall be protected against backflow from the premises by installing an approved backflow prevention assembly in the service line appropriate to the degree of hazard. This shall include the handling of process waters and waters originating from the utility system which have been subjected to deterioration in quality.

3.2.3.3 In the case of premises having (1) an internal cross connection that cannot be permanently corrected or controlled, or (2) intricate plumbing and piping arrangements or where entry to all portions of the premises is not readily accessible for inspection purposes, making it impracticable or impossible to ascertain whether or not dangerous cross connections exist, the public or nonpublic water system shall be protected against backflow from the premises by installing an approved backflow prevention assembly in the service line.

3.2.4 The type of protective assembly required under subsections 3.2.3.1 - 3.2.3.3 shall depend upon the degree of hazard which exists as follows:

3.2.4.1 In the case of any premises where there is an auxiliary water supply as stated in subsection 3.2.3.1 of this section and it is not subject to regulation by the National Park Service, the public or nonpublic water system shall be protected by an approved air gap separation or an approved reduced pressure principle backflow prevention assembly.

3.2.4.2 In the case of any premises where there is water or a substance(s) that would be objectionable but not hazardous to health if introduced into the public or nonpublic water system, the public or nonpublic water system shall be protected by an approved double check valve assembly.

3.2.4.3 In any case of any premises where there are “uncontrolled” cross connections, either actual or potential, the public or nonpublic water system shall be protected by an approved air gap separation or an approved reduced pressure principle backflow prevention assembly at the service connection.

3.2.5 Any backflow prevention assembly required herein shall be a model and size approved by the **[Park Position - Chief of Maintenance./Facilities Management suggested here]**. The term “approved backflow assembly” means an assembly that has been manufactured in full conformance with the latent standard established by the American Water Works Association entitled:

AWWA C506 Standards for Reduced Pressure Principle and Double Check Valve Backflow Prevention Device and, has met the laboratory and field performance specifications of the Foundation for Cross Connection Control and Hydraulic Research of the University of Southern California established by:

Specifications of Backflow Prevention Assemblies - Section 10 of the most current issue of the Manual of Cross Connection Control.

Said AWWA and FCCC&HR standards and specifications have been adopted by the **[Park Position - Chief of Maintenance./Facilities Management suggested here]**. Final approval shall be evidenced by a “Certificate of Approval” issued by an approved testing laboratory certifying full compliance with the said AWWA standards and FCCC&HR specifications.

The following testing laboratory has been qualified by the **[Park Position - Chief of Maintenance./Facilities Management suggested here]** to test and certify backflow preventers:

Foundation for Cross Connection Control and Hydraulic Research
University of Southern California
University Park
Los Angeles, CA 90089

Testing laboratories other than the laboratory listed above will be added to an approved list as they are qualified by the **[Park Position - Chief of Maintenance./Facilities Management suggested here]**

Backflow preventers, which may be subjected to backpressure or backsiphonage, that have been fully tested and have been granted a Certificate of Approval by said qualified laboratory and are listed on the laboratory’s current list of “Approved backflow prevention assemblies”, may be used without further qualification. However, annual testing by a state certified backflow prevention specialist is required to ensure proper continual operation.

3.2.6 It shall be the duty of the building owner at any premises where backflow prevention assemblies are installed to have certified inspections and operational tests made upon installation and at least once per year. In those instances where the **[Park Position - Chief of Maintenance./Facilities Management suggested here]** deems the hazard to be great enough he may require certified inspectors at more frequent intervals. These inspections and tests shall be at the expense of the building owner and shall be performed by a state certified tester. It shall

be the duty of the **[Park Position - Chief of Maintenance./Facilities Management suggested here]** to see that these tests are made in a timely manner. The building owner shall notify the **[Park Position - Chief of Maintenance./Facilities Management suggested here]** in advance when the tests are to be undertaken so that an official representative may witness the tests if so desired. These assemblies shall be repaired, overhauled or replaced at the expense of the building owner whenever said assemblies are found to be defective. Records of the such tests, repairs and overhaul shall be kept and made available to the **[Park Position - Chief of Maintenance./Facilities Management suggested here]** or the Regional Public Health Consultant.

- 3.2.7** All presently installed backflow assemblies which do not meet the requirements of this section but were approved assemblies for the purpose described herein at the time of installation and which have been properly maintained, shall, except for the inspection and maintenance requirements under subsection 3.2.6, be excluded from the requirements of these rules so long as the **[Park Position - Chief of Maintenance./Facilities Management suggested here]** and the Regional Public Health Consultant are assured that they will satisfactorily protect the utility system. Whenever the existing assembly is moved from its present location or requires more than the minimum maintenance or when the **[Park Position - Chief of Maintenance./Facilities Management suggested here]** or Regional Public Health Consultant finds that the maintenance constitutes a hazard to health, the unit shall be replaced by an approved backflow prevention assembly meeting all the requirements of this section.

Date:

State Certified Cross Connection Control Program Specialists

Last Name	First Name	Certification Number	Certification Expiration Date
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